Economics 508: Problem Set M

– How to determine the monopoly price in a stochastic environment

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Monopolist's Problem

- The monopolist wants to maximize the present value of his expected future flow of profit.

$$\max_{[p_t]_{t=0,1,2,\ldots}} \sum_{t=0}^{\infty} \beta^t E[\pi(p_t)|I_t]$$

(When there is no discount factor, then $\beta = 1$ and when there is a discount factor, then $\beta < 1$)

- Let's assume that the # of trials for the price experiment is limited to n. Then the monopolist will solve:

$$\max_{[p_t]_{t=0,1,\ldots n}} \left[\sum_{t=0}^{n-1} \beta^t E[\pi(p_t)|I_t] + \sum_{t=n}^{\infty} \beta^t E[\pi(p^*)|I_n] \right]$$

When there is no discount factor

- We should find a sequence of price which maximizes $(X^T X)$, where $X = [1 \ \vdots \ \log(p) \ \vdots \ \log^2(p)]$ (so that the variance of estimator $\sigma^2 (X^T X)^{-1}$ have minimum variance).

  The approximate efficient choice of $\{p_t\}$ can be thus

Strategy I: $[p'_t]_{t=0,1,\ldots,n-1} = [\underline{p}', \underline{p}' + \frac{\overline{p}' - \underline{p}'}{n-1}, \underline{p}' + 2 \cdot \frac{\overline{p}' - \underline{p}'}{n-1}, \ldots, \overline{p}']$

where p_t' are log transform of prices

- For example, in $n = 10$, $p_t \in [\underline{p}, \overline{p}] = [0, 60] \ \forall t = 0,1,2,\ldots$, case, our choice of price sequence can be:

  $[p_t]_{t=0,1,\ldots,9} = [1.6 \ 2.5 \ 3.7 \ 5.5 \ 8.2 \ 12.2 \ 18.2 \ 27.1 \ 40.4 \ 60.3]$

When there is a discount factor

- We can make use of the sequence
  $[p_t]_{t=0,1,\ldots,9} = [1.6\ 2.5\ 3.7\ 5.5\ 8.2\ 12.2\ 18.2\ 27.1\ 40.4\ 60.3]$

- We may just reorder the sequence so that we can maximize the discounted profit flow during the experimental period.

Strategy I ':

① At time 0, choose an arbitrary $\log(p_0) = 1$, get $\pi(p_0)$.

② At time 1, choose an arbitrary $\log(p_1) = 4$, get $\pi(p_1)$.

③ At time 2, if $\dfrac{\pi(p_1) - \pi(p_0)}{p_1 - p_0} \gg 0$, choose $\log(p_2) = 1.5$

if $\dfrac{\pi(p_1) - \pi(p_0)}{p_1 - p_0} \ll 0$, choose $\log(p_2) = 4.5$, and

Strategy I ' cont'

if $sign(\frac{\pi(p_1)-\pi(p_0)}{p_1-p_0})$ is ambiguous, choose $p_2 = 2.5$

④ At this point, we have 3 pairs of price and demand, thus we can estimate the demand curve.

⑤ Using this demand curve and cost function, we can find a preliminary p^*, and let $p_3 =$ the price sequence closest to p^* among the left over price sequences $\{p_t\}$.

⑥ Again, get another pair of price and demand, reestimate the demand curve, and get a new p^* from this demand curve. Choose $p_4 =$ the price sequence closest to p^* among the left over price sequences.

⑦ Repeat the same procedure until $t = 9$. and the p^* from the last demand eqution becomes our monopolist price.

Strategy II (Cauchy's Method)

- *Strategy I'* guarantees same expected accuracy of profit maximizing price and a better profit stream than Strategy I.

- However, we need to maximize the lifetime expected profit stream, thus we need to estimate profit maximizing price, and at the same time we should reach there as fast as we can.

Strategy II (Cauchy's Method):

① At time 0, choose an arbitrary p_0, get $\pi(p_0)$.

② At time 1, choose an arbitrary p_1, get $\pi(p_1)$.

③ At time 2, choose an arbitrary s_2, a step size.

Strategy II (Cauchy's Method) cont'

④ $p_t = p_{t-1} + s_t \cdot u_t$,

where $w_t = \dfrac{\pi(p_{t-1}) - \pi(p_{t-2})}{p_{t-1} - p_{t-2}}$

$u_t = w_t / |w_t|$, thus $u_t = \begin{cases} +1 \\ -1 \end{cases}$ (a directional component)

and $s_t = \left(\dfrac{w_t - w_{t-1}}{p_t - p_{t-1}}\right)^{-1}$ (a step size) $\forall\ t = 2, 3, \ldots$

Strategy II' (Modified Method)

- Through this method, he can reduce the number of the experimental trial of setting price level too much far away from the profit maximizing price level, and ascend to the maximum profit via steepest route.

- However, when it happens to have approximately the same value of slope at time t−1 and t−2, i.e., $w_{t-1} = w_{t-2}$, the denominator of s_t becomes close to 0 and the price process takes too much a big step towards the optimum value.

- Since the monopolist know the right form of the demand curve, he can just guess the demand curve with his experimental trials.

Strategy II' (Modified Method):

① At time 0, choose an arbitrary p_0, get $D(p_0)$ and associated $C(D(p_0))$.

② At time 1, choose an arbitrary p_1, get $D(p_1)$ and associated $C(D(p_1))$.

Strategy II' (Modified Method) cont'

③ At time 2, choose an arbitrary s, a step size.

④ $p_3 = p_2 + s \cdot u$, where the definition of u is the same as u_2.

⑤ At this point, we have 3 pairs of price and demand, thus we can estimate the demand curve.

⑥ Using this demand curve and cost function, we can find a preliminary p^*, and let $p_4 = p^*$

⑦ Again, get the demand for p_4, get another pair of price and demand, reestimate the demand curve. p^* from this demand curve becomes a new price strategy for the next period.

⑧ Repeat the same procedure for n times, and the last n-th p^* becomes his choice of monopolist price afterwards.

Optimum Strategy

- Strategy I : Choose a sequence of price with maximum variance during the experimental period
- Strategy II' : Try to choose the best guess of profit maximizing price during the experimental period

- Thus,

$$[\sum_{t=0}^{n-1} E\pi(p_t) \mid Strategy\ \mathrm{I}\ ] < [\sum_{t=0}^{n-1} E\pi(p_t) \mid Strategy\ \mathrm{II}'\ ]$$

$$[\sum_{t=n}^{\infty} E\pi(p_t) \mid Strategy\ \mathrm{I}\ ] > [\sum_{t=n}^{\infty} E\pi(p_t) \mid Strategy\ \mathrm{II}'\ ]$$

Simulation

Some Assumptions Regarding the Simulation:

Assumption 1 : $p_t \in [\underline{p}, \overline{p}] = [0, 50] \ \forall t = 0, 1, 2, \ldots$

Assumption 2 : The monopolist can bear only 10 experimental trial of his price. After that, he has to choose one fixed monopolist price.

Assumption 3 : $\beta = 0.92$

Simulations are for the cases where:

Case A. Use Strategy I, i.e., $[\log(p_t)]_{t=0,1,\ldots,9} = [0.5 \ 0.9 \ 1.3 \ 1.7 \ 2.1 \ 2.5 \ 2.9 \ 3.3 \ 3.7 \ 4.1]$

Case B. Use Strategy II' with $p_0 = 30, p_1 = 20, s = 5$

Case C. Use Strategy II' with $p_0 = 50, p_1 = 40, s = 5$

Case D. Use Strategy II' with $p_0 = 30, p_1 = 20, s = 10$

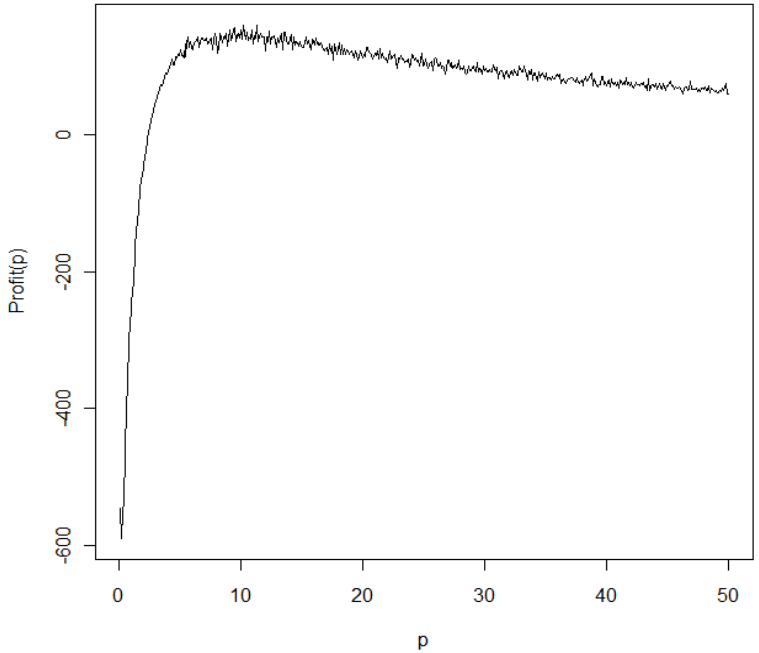
Simulation Results

		p^*	Profit Stream $\sum_{t=0}^{9} \beta^t \pi(p_t)$	Profit Stream $\sum_{t=10}^{100} \beta^t \pi(p_t)$	Estimated Demand Curve ($F_{2,7}$ statistics)
	1st Trial	9.32	479.85	828.69	$4.91-0.41\log(p)-0.19\log^2(p)$ (2537)
Case A	2nd Trial	9.30	455.92	804.73	$5.07-0.60\log(p)-0.15\log^2(p)$ (17190)
	3rd Trial	9.44	482.18	831.60	$4.92-0.43\log(p)-0.19\log^2(p)$ (6860)
Case B		8.49	915.00	776.40	$5.35-0.77\log(p)-0.13\log^2(p)$ (1174)
Case C		8.40	769.39	771.89	$5.28-0.68\log(p)-0.15\log^2(p)$ (3054)
Case D		10.03	916.03	771.49	$4.48-0.14\log(p)-0.24\log^2(p)$ (420)

Choice of the Price Process for the first periods

t	0	1	2	3	4
Case A	1.60	2.50	3.70	5.50	8.20
Case B	30.00	20.00	15.00	5.32	8.70
Case C	50.00	40.00	35.00	32.57	8.38
Case D	30.00	20.00	10.00	7.72	10.43

t	5	6	7	8	9	10
Case A	12.20	18.20	27.10	40.40	60.30	9.32
Case B	8.54	8.49	8.50	8.47	8.45	8.49
Case C	8.38	8.26	8.16	8.49	8.44	8.40
Case D	10.17	10.13	10.34	10.38	10.03	10.03



Compare the performances: Strategy I, Strategy II' and KWopt

	Profit Stream		Optimum Price Estimation
	$\sum_{t=0}^{9} \beta^t \pi(p_t)$	$\sum_{t=10}^{100} \beta^t \pi(p_t)$	p^*
KW opt	Middle	Middle	Somewhat Accurate
Strategy I	Small	Big	Accurate
Strategy II'	Big	Small	Inaccurate

- For the same numbers of trial ($n = 10$), KW optimization algorithm is somewhere between Strategy I and II'.